ET AI HACKATHON 2.0 · FINALE · PROBLEM STATEMENT 5

VayuNetra

The operations layer for urban air quality.

We don't just measure the air. We **trace** it, **predict** it, and **act** on it — live in ten Indian cities, on ₹0 infrastructure.

DECEMBER 2025 · WORST DAY OF THE MONTH

452	345	331	302
Delhi	**Chennai**	**Kolkata**	**Pune**
Severe	Very Poor	Very Poor	Very Poor

Indian National AQI from each city's own CPCB stations. 10 of 10 cities have a December 2025 record.

Team DaGoats · Omkar Kadam · Sejal Kumbhar · Abhinav Prasad

Live: vayunetra-aqi.vercel.app · dots = worst day in December 2025 · built 19 Aug 2026

THE PROBLEM

The data exists. The layer that acts on it doesn't.

1.67M

premature deaths every year in India from air pollution

Lancet Planetary Health (GBD), as cited in the brief

900+

CAAQMS monitoring stations already deployed — the data exists

CPCB continuous monitoring network

31%

of monitored cities have **any** response protocol

CAG performance audit, 2024

A reading turns red. Then nothing happens. No system tells a city officer **who is polluting this square kilometre right now**, **what the air will be tomorrow**, and **where to send inspectors first**.

Status quo

reactive orders — 13 of 17 GRAP orders last winter came *after* the AQI had crossed the threshold; weeks to enforcement

VayuNetra

cited recommendation in **seconds**; approval → dispatch → closure timestamped on an audit trail

WHAT VAYUNETRA DOES

One loop: trace → predict → act → protect

TRACE

Who is to blame, per ~1 km² cell — GBM + SHAP blended with chemical-signature priors; abstains where it lacks skill.

Delhi, latest run: transported 29% · traffic 28% · other 19.3%

→

PREDICT

PM2.5 at +24/48/72 h per cell, 80 % conformal band, calibrated P(>120), P(>250).

share of Delhi expected in Very Poor air (calibrated P)

from calibrated P(>120) per cell × population · GET /exposure

→

ACT

Ranked worklist → Sentinel-2 evidence + retrieved regulation → draft notice → approve · dispatch · **close with finding**, on an audit trail.

413

ranked, cited recommendations across ten cities · draft notice PDF each

live worklist, this morning

→

PROTECT

Templated advisories (no LLM) in eight scripts — app, Telegram, IVR line, public displays.

Delhi Ward 12: air is forecast very poor in +24h. Keep outdoor activity short, use an N95 mask, and move heavy work outside the peak window.

English · Very Poor, +24 h · templated

Six AI agents on one graph — signal → cited recommendation in **0.9 s** of compute, measured live; approval, dispatch and closure timestamped per action. The spatial unit is the same ~1 km cell everywhere, so every claim traces from a raw reading to an action.

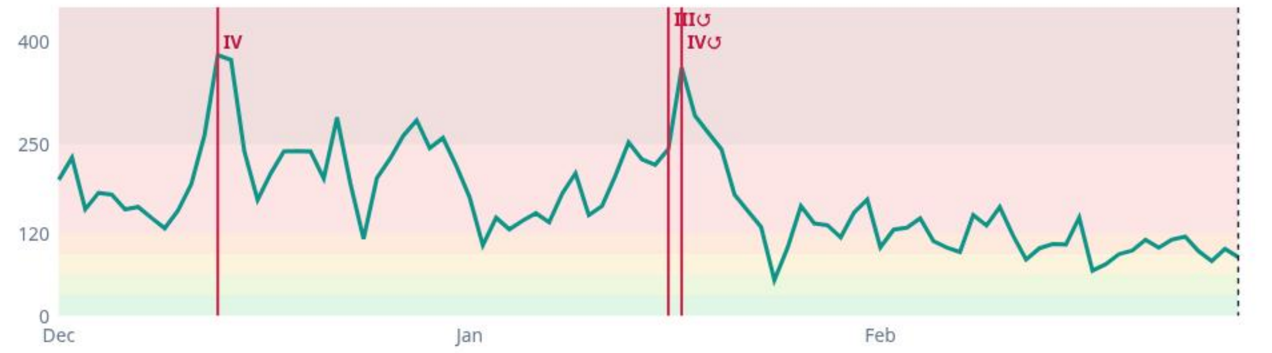
EARLY WARNING, IN HINDSIGHT

Delhi's winter, replayed — against the government's real orders

216.3 **Dec 2025** mean µg/m³ · AQI 374 Very Poor **175.3** **Jan 2026** mean µg/m³ · AQI 343 Very Poor **106.6** **Feb 2026** mean µg/m³ · AQI 254 Poor

39 station cells (drawn 2.2× for legibility) · daily mean PM2.5 · CPCB via OpenAQ
outline = Delhi municipal wards; some CPCB stations sit outside it, in the wider NCR

city-mean PM2.5 (µg/m³) · daily **28 Feb 2026**



Stage IV 2025-12-13
24 h before: **P(>120) 83%** · 99% of 38 cells
persistence said 225 · at order 382

Stage III ↺ 2026-01-16
24 h before: **P(>120) 85%** · 100% of 1 cells ⚠
persistence said 205 · at order 214

Stage IV ↺ 2026-01-17
24 h before: **P(>120) 72%** · 100% of 1 cells ⚠
persistence said 131 · at order 333

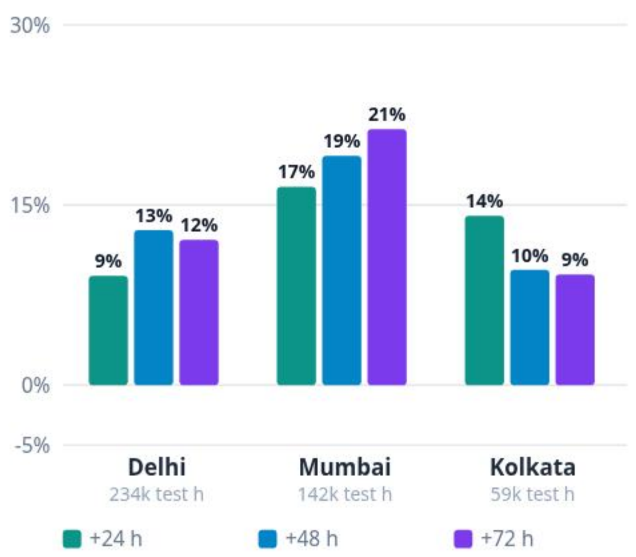
Served forecast replayed with the production recipe (rolling monthly refit, persistence blend, calibrated P). Green = P(>120) ≥ 30 % a day before the order. ⚠ = the public feed carried a single Delhi station that week. Sources: CAQM / PIB releases, linked in docs/OUTCOMES.md.

TECHNICAL RIGOUR

Every number is measured — including our failures

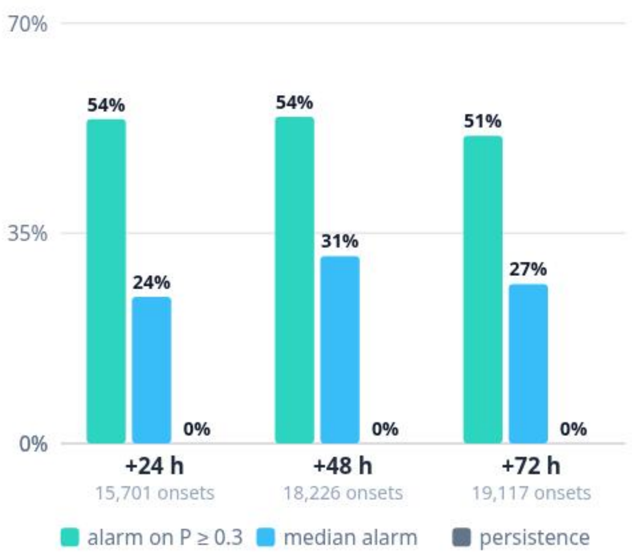
Served forecast skill vs persistence

1 − RMSE / $RMSE_{persistence}$ · strict temporal split, monthly refit on trailing 90 d



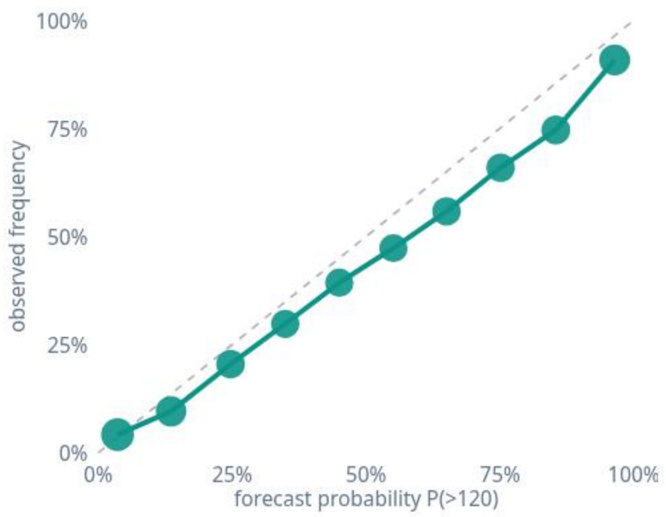
Warns before the air turns

clean → Very-Poor onsets flagged, alarm on calibrated P ≥ 0.3 · Delhi, multi-season



The probabilities are honest

reliability of P(>120) at +24 h · 80 % band coverage



- Attribution vs published apportionment: cosine 0.88 / 0.90 / 0.93 · bucket tables vs TERI-ARAI 2018 & Guttikunda 2019 — disagreements stated
- TFT trained on GPU and rejected — LightGBM won held-out skill
- 204 backend tests · 16 e2e flows · CI on every push

LIVE DEMO

Delhi, right now.

1 **Morning brief → blame map**
what changed overnight, where the air turns; click the worst hexagon → sources, SHAP, R^2 gate, 72-h forecast with P(Very Poor)

2 **Worklist → evidence → notice**
real Sentinel-2 image + retrieved regulation → draft notice PDF; Approve · Dispatch · **Close case** · History

3 **Advisories in eight scripts**
Hindi → Tamil; live Telegram bot; IVR line

4 **Proof, then ten cities**
validation card · real orders in hindsight · scoreboard → Mumbai

press D opens the live console inside this deck · Esc returns · offline: the last screenshot is shown, labelled

BUSINESS IMPACT · SCALABILITY

Built to deploy, not to demo

What it costs to run — measured, not assumed

monthly infrastructure vs cities · one row per reading · 180-day raw retention with archive

Supabase free 500 MB → Pro US$25; Render free → Starter US$7 for an always-on API. docs/SCALE.md

RUNNING AT PRODUCTION SCALE — LIVE

10
cities live · Delhi to Lucknow

16,529
~1 km² cells modelled

647
registered + satellite-detected sources

5,495
vulnerability-scored zones

413
RAG-cited enforcement recommendations

8
advisory scripts · app / Telegram / IVR / displays

Adoption path

- **One YAML per city** (bbox, languages, authority) + one backfill run — 7 metros in a week
- **Closed loop**: approve → dispatch → close with finding, immutable audit trail
- **PRANA-ready export** of every tracked action with its measured effect — feeds the official NCAP portal
- Delhi ROI card: **₹0.91 L cr/yr of Delhi's annual health burden avertable at the ~30 % NCAP target (18,219 deaths/yr) — every figure cited**

India measures. India forecasts.

VayuNetra operates.

Traced, predicted, cited, delivered, **closed**. In seconds. In eight scripts. For ₹0.

vayunetra-aqi.vercel.app

github.com/omkarrr88/VayuNetra · @aqivayu_bot

Omkar Kadam · Sejal Kumbhar · Abhinav Prasad — thank you.

APPENDIX

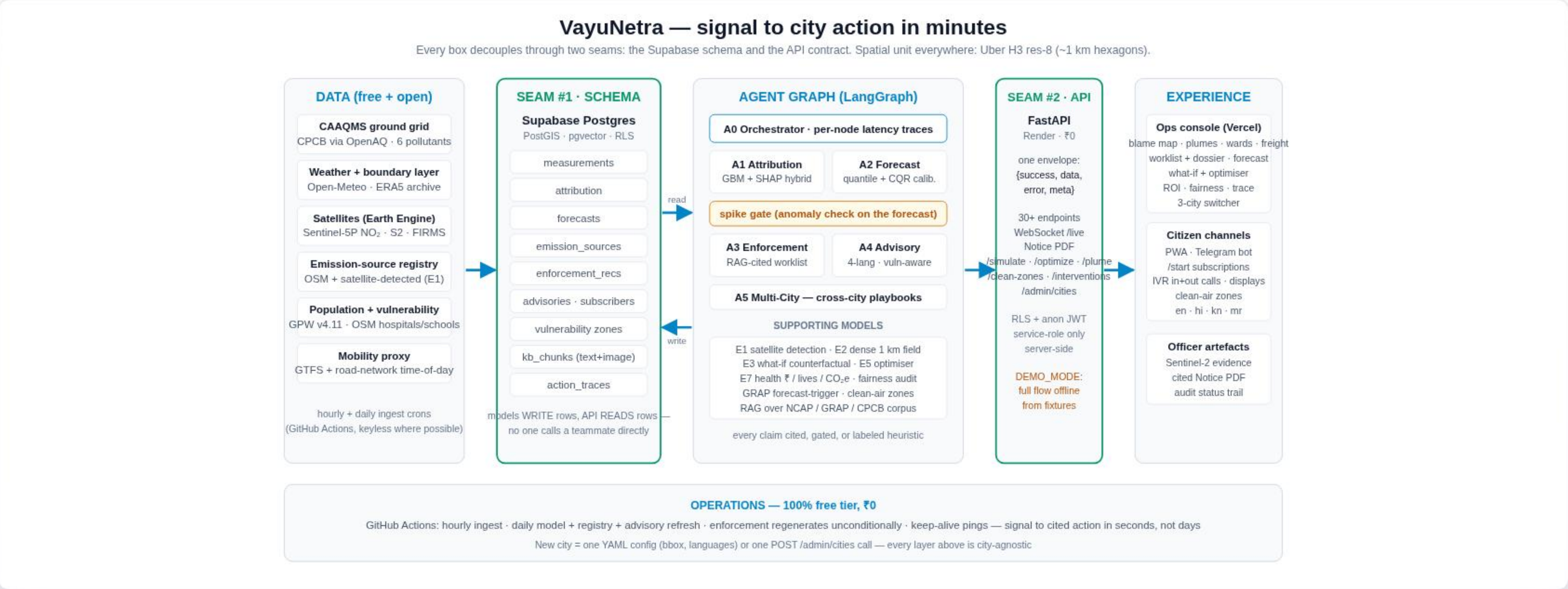
For the Q&A

A1 Architecture
A2 Six agents live
A3 Evidence dossier
A4 Citizen reach — eight scripts
A5 Models & validation, all ten cities

A6 Honesty & fairness
A7 Cost, security & roadmap
A8 Ten cities
A9 Past air
A10 Positioning vs SAFAR / DSS / PRANA

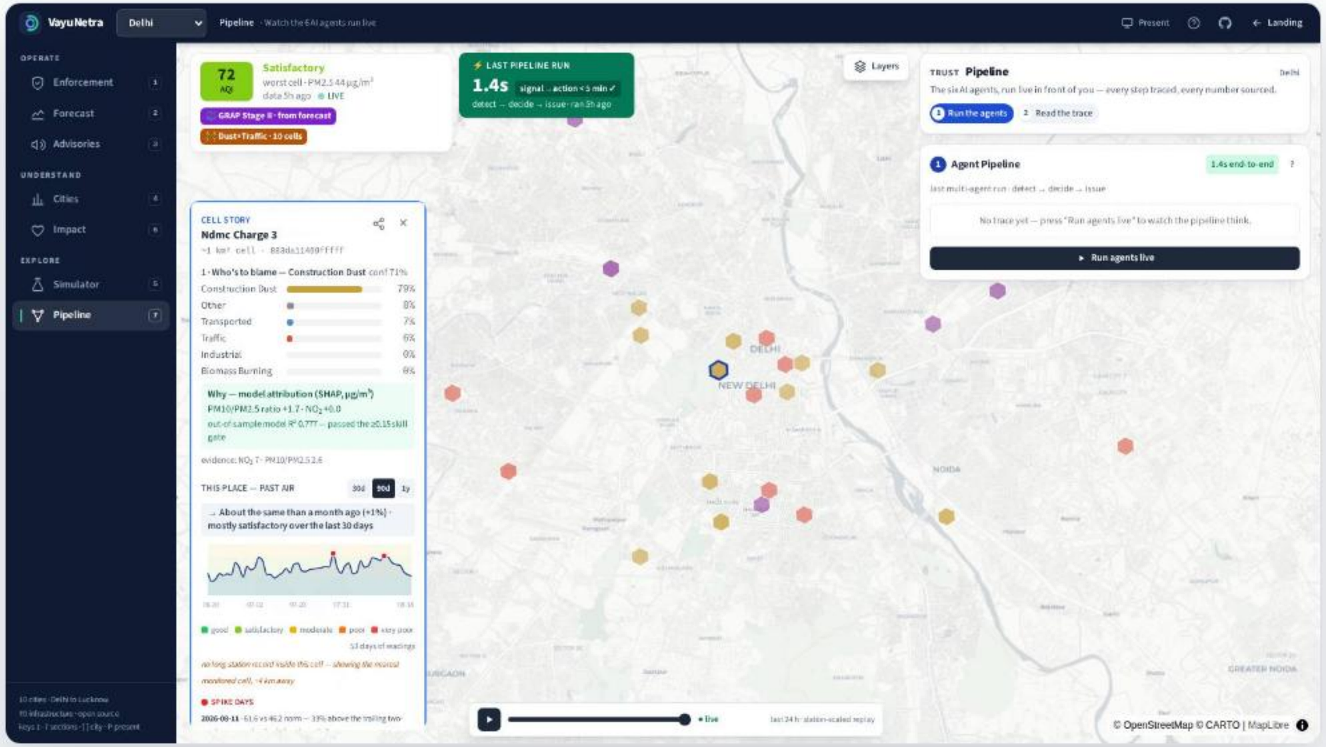
A11 Early warning & calibration
A12 Attribution validation & exposure
A13 Real orders in hindsight — did the air change?
A14 The officer loop closes — audit trail

Signal to cited action — two seams, one envelope



FastAPI (44 routes + WebSocket) on Render · React + MapLibre / deck.gl on Vercel · Supabase Postgres + PostGIS + pgvector with RLS (writes only through the service role) · LangGraph agents · GitHub Actions crons — all free tier for the ten cities running today.

Six agents on one graph, traced per node



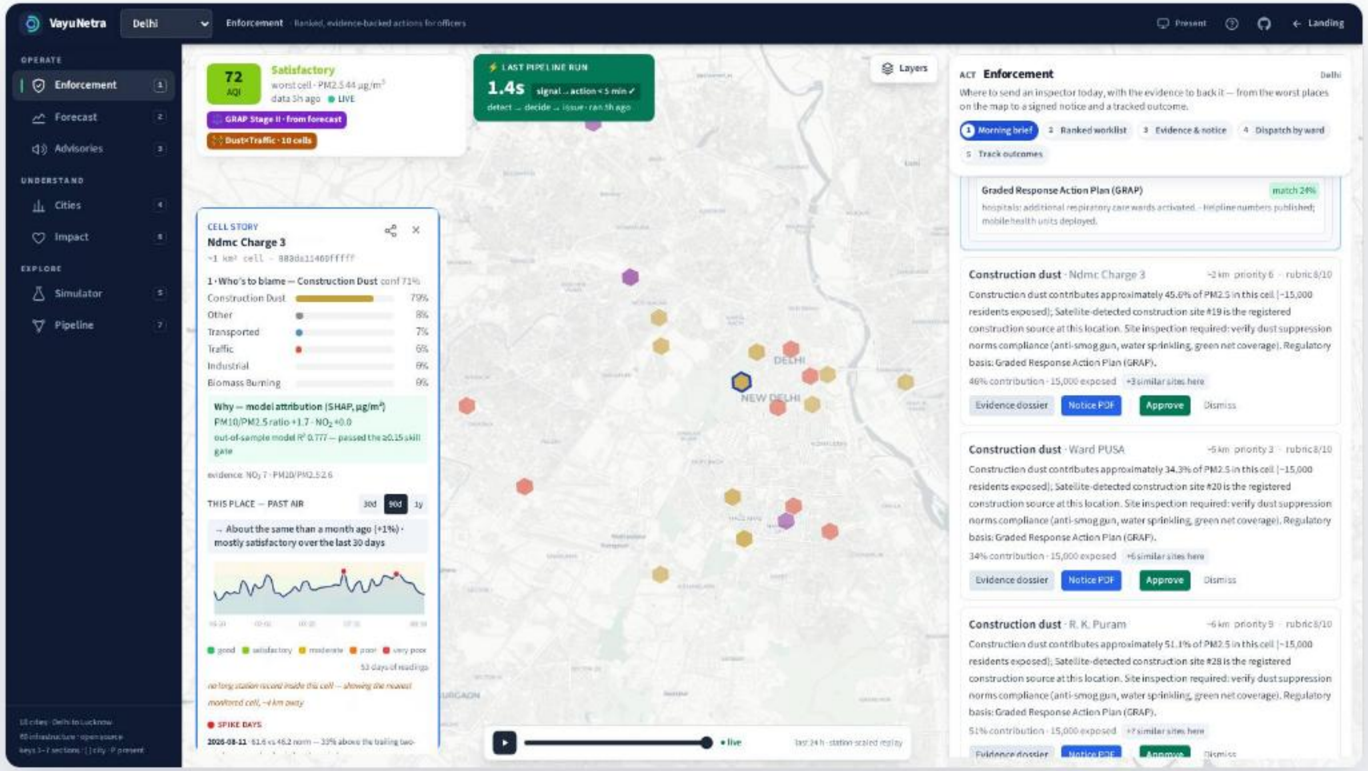
Latest trace, Delhi

attribution agent	**282 ms**
forecast agent	**298 ms**
enforcement agent — skipped this run (spike gate: no spike)	**–**
advisory agent	**287 ms**
signal → cited recommendation	**0.9 s**

action_traces · signal → cited recommendation is compute time; approval / dispatch / closure are separate timestamped steps

A3 · EVIDENCE

A recommendation you can defend



Real Sentinel-2 imagery of the actual site (multimodal RAG over image patches + regulatory text).

Regulation retrieved verbatim — cited chunk by chunk; the right instrument for the city (GRAP only in Delhi-NCR).

Draft notice PDF: addressee, provisions, 24-h deadline, satellite image, projected impact of compliance (screening estimate), provenance block. Stamped **pending officer authorisation**.

A4 · CITIZENS

The same advisory, in the reader's own script

ENGLISH

Delhi Ward 12: air is forecast very poor in +24h. Keep outdoor activity short, use an N95 mask, and move heavy work outside the peak window.

HINDI

Delhi Ward 12: अगले 24 घंटों में हवा बहुत ख़राब रहने का अनुमान है. बाहर की गतिविधि कम रखें, N95 मास्क पहनें, और भारी काम पीक समय के बाद करें.

KANNADA

Delhi Ward 12: ಮುಂದಿನ 24 ಗಂಟೆಗಳಲ್ಲಿ ಗಾಳಿಯ ಗುಣಮಟ್ಟ ತುಂಬಾ ಕಳಪೆ ಇರಲಿದೆ ಎಂದು ಅಂದಾಜಿಸಲಾಗಿದೆ. ಹೊರಗಿನ ಚಟುವಟಿಕೆ ಕಡಿಮೆ ಮಾಡಿ, N95 ಮಾಸ್ಕ್ ಬಳಸಿ, ಮತ್ತು ಭಾರೀ ಕೆಲಸವನ್ನು ಪೀಕ್ ಸಮಯದ ನಂತರ ಮಾಡಿ.

MARATHI

Delhi Ward 12: पुढील 24 तासांत हवा खूप खराब राहण्याचा अंदाज आहे. बाहेरील हालचाल कमी ठेवा, N95 मास्क वापरा, आणि जड काम पीक वेळेनंतर करा.

TAMIL

Delhi Ward 12: அடுத்த 24 மணி நேரத்தில் காற்றின் தரம் மிக மோசம் இருக்கும் என எதிர்பார்க்கப்படுகிறது. வெளியில் செல்வதைக் குறைக்கவும், N95 முகக்கவசம் அணியவும், கடின வேலைகளை உச்ச நேரத்திற்குப் பிறகு செய்யவும்.

TELUGU

Delhi Ward 12: రాబోయే 24 గంటల్లో గాలి నాణ్యత చాలా చెడ్డదిగా ఉండవచ్చు. బయటి కార్యకలాపాలు తగ్గించండి, N95 మాస్క్ ధరించండి, భారీ పనిని పీక్ సమయం తర్వాత చేయండి.

BENGALI

Delhi Ward 12: আগামী 24 ঘণ্টায় বাতাস খুব খারাপ থাকার পূর্বাভাস। বাইরে কম সময় থাকুন, N95 মাস্ক পরুন এবং ভারী কাজ পিক সময়ের পরে করুন।

GUJARATI

Delhi Ward 12: આગામી 24 કલાકમાં હવા ખૂબ ખરાબ રહેવાની આગાહી છે. બહારની પ્રવૃત્તિ ઓછી રાખો, N95 માસ્ક પહેરો અને ભારે કામ પીક સમય પછી કરો.

Deterministic templates (no LLM), script-validated; app · Telegram bot · IVR line (Hindi voice for Hindi-first cities) · public displays; 5,495 vulnerability-scored zones (hospitals, schools, elder-care, outdoor work × population).

All ten cities, both baselines — negatives kept

city · window	test hours @24 h	vs persistence 24/48/72 h	vs weekly seasonal-naive	80 % PI coverage
Delhi · multi-season	2,34,491	+9% / +13% / +12%	+26% / +20% / +16%	78% / 78% / 77%
Mumbai · multi-season	1,41,667	+17% / +19% / +21%	+37% / +35% / +32%	82% / 82% / 79%
Kolkata · multi-season	59,373	+14% / +10% / +9%	+26% / +14% / +13%	75% / 73% / 70%
Ahmedabad · live 90 d	745	+14% / +13% / +3%	+19% / +13% / +1%	79% / 85% / 74%
Bengaluru · live 90 d	1,127	+14% / +42% / +47%	+11% / -8% / +3%	83% / 83% / 87%
Chennai · live 90 d	884	+6% / +5% / +6%	+22% / +22% / +22%	79% / 75% / 75%
Delhi · live 90 d	2,235	+16% / +19% / +12%	+27% / +22% / +19%	74% / 74% / 72%
Hyderabad · live 90 d	760	+11% / +25% / +37%	+32% / +31% / +27%	78% / 81% / 73%
Jaipur · live 90 d	941	+6% / -4% / -6%	+26% / +8% / +3%	85% / 82% / 70%
Kolkata · live 90 d	354	+12% / +12% / +30%	+28% / +36% / +28%	82% / 78% / 74%
Lucknow · live 90 d	810	+26% / +24% / +23%	+73% / +33% / +32%	81% / 79% / 75%
Mumbai · live 90 d	1,877	+10% / +14% / +6%	+51% / +51% / +30%	72% / 70% / 80%
Pune · live 90 d	1,297	+20% / +21% / +32%	+25% / +24% / +28%	73% / 78% / 84%

Protocol: strict temporal split, rolling monthly refit on the trailing 90 days (= production), one shared support mask; served forecast = LightGBM median blended with persistence, conformal 80 % band, calibrated exceedance probabilities. TFT-lite trained on GPU and rejected. python -m ml.eval.benchmark · GET /metrics/benchmark

A6 · HONESTY & FAIRNESS

Honest by construction

The model abstains

No ML blame without out-of-sample skill ($R^2 \geq 0.15$); otherwise cited chemical-signature priors, and the UI says which.

Health text is LLM-free

Deterministic templates in eight scripts — a language model cannot hallucinate medical advice.

Notices are drafts

Stamped pending officer authorisation; never auto-sent. Dispatch starts before/after measurement against the city's drift.

Nothing fabricated

Impact figures return null over invented constants; satellite evidence is real imagery of the real site; study numbers are read from the primary PDFs.

No socio-economic inputs

Priority = contribution × exposure × actionability × confidence. No income or land-value feature anywhere in the pipeline.

Auditable end to end

Per-node agent traces + an immutable status log (who, when, from → to, finding) that outlives the record it describes.

₹0 today — a config file per city, a measured path forward

Stack · ₹0

FastAPI on Render · React + MapLibre on Vercel · Supabase Postgres (PostGIS + pgvector) · LangGraph · GitHub Actions crons. Free tier for the ten cities running today.

Where ₹0 stops

cities	DB @180 d	₹ / month
10 (today)	~465 MB	0
15	~655 MB	~2,100
50	~2.0 GB	~2,100
131 (all NCAP)	~5.1 GB	**~2,700**

measured 0.21 MB/city/day; one row per reading; archive-first retention

Security & roadmap

RLS on every table; public reads via anon JWT; writes only through the service role on the server; admin onboarding behind a key header. Roadmap: InMAP/PAVITRA source–receptor physics for the what-if engine, a learned satellite CV detector, more in-language IVR voices, permit-registry connectors.

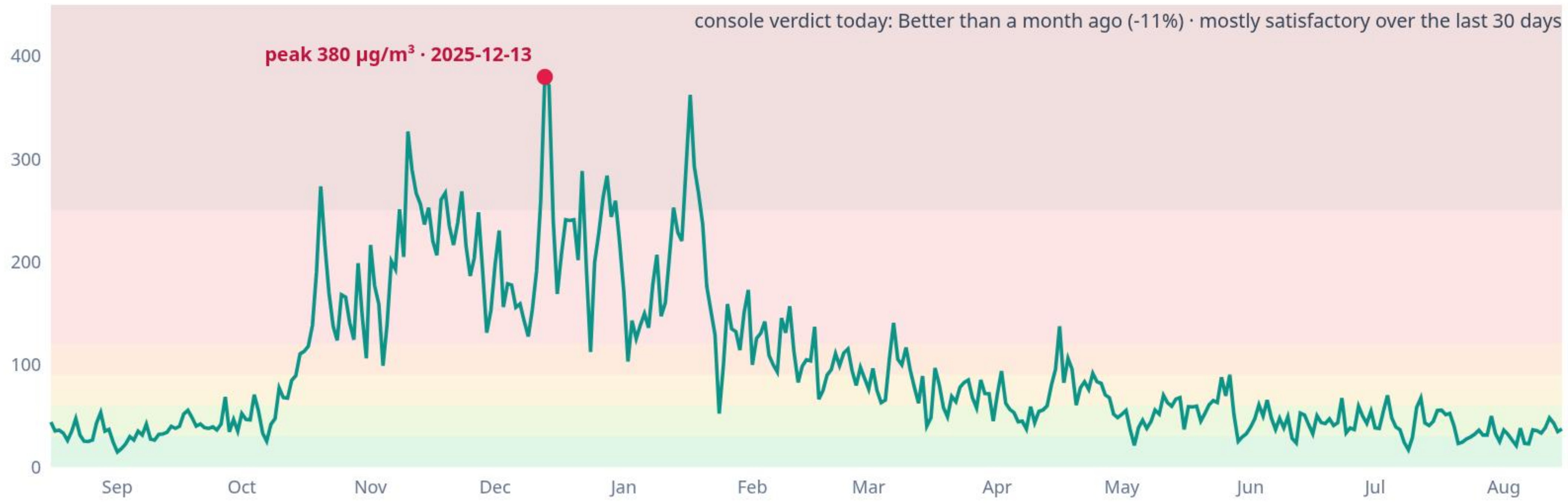
A8 · TEN CITIES

One engine, ten cities — live

#	city	PM2.5 now	+24 h	dominant source	languages
1	**Jaipur**	39.2	34.2	transported	hi en
2	**Ahmedabad**	35.1	25.6	transported	gu en hi
3	**Delhi**	34.2	32.2	transported	hi en
4	**Kolkata**	25.1	22.5	construction dust	bn en hi
5	**Hyderabad**	21.7	18.6	industrial	te en hi
6	**Lucknow**	21.7	19.0	industrial	hi en
7	**Chennai**	19.7	21.5	construction dust	ta en
8	**Bengaluru**	19.2	11.4	industrial	kn en
9	**Pune**	19.0	16.0	construction dust	mr en
10	**Mumbai**	16.1	11.8	construction dust	mr en hi

live comparison · as of 19 Aug, 11:33 am

Delhi, one year of daily PM2.5 — the real winter



station-day means over 39 Delhi stations (CPCB via OpenAQ) — the same record the forecast benchmark uses; CPCB bands · the console shows the same view from its live table ∪ archived rollup (GET /history/trend)

India can measure and forecast its air. The gap is acting on it — audited.

What exists

SAFAR (IITM, 4 cities, city-level bulletin) · AQEWS / DSS (WRF-Chem, 400 m grid, 29-source — CAQM/government users only) · PRANA (cities report NCAP actions & spend) · NCAP Tracker · CAAQMS / SAMEER (a reading, not a decision).

What the audit found — CEEW, Oct 2025

'Severe and above' caught 5 of 14 episodes (1 of 15 the winter before) · PM2.5 under-predicted by ~24 μg/m³ in winter · CAQM imposed GRAP III/IV on *observed* AQI, not on forecasts · DSS lacks outcome monitoring and actionable short-term pathways.

What VayuNetra adds

Calibrated P(>band) on every cell · onset recall reported vs persistence · live hourly attribution, not a 2016 inventory · forecast → named notice with citations → dispatch → measured outcome, closed with a finding · public, 8 languages, IVR/Telegram/web · ₹0 on the sunk CAAQMS investment.

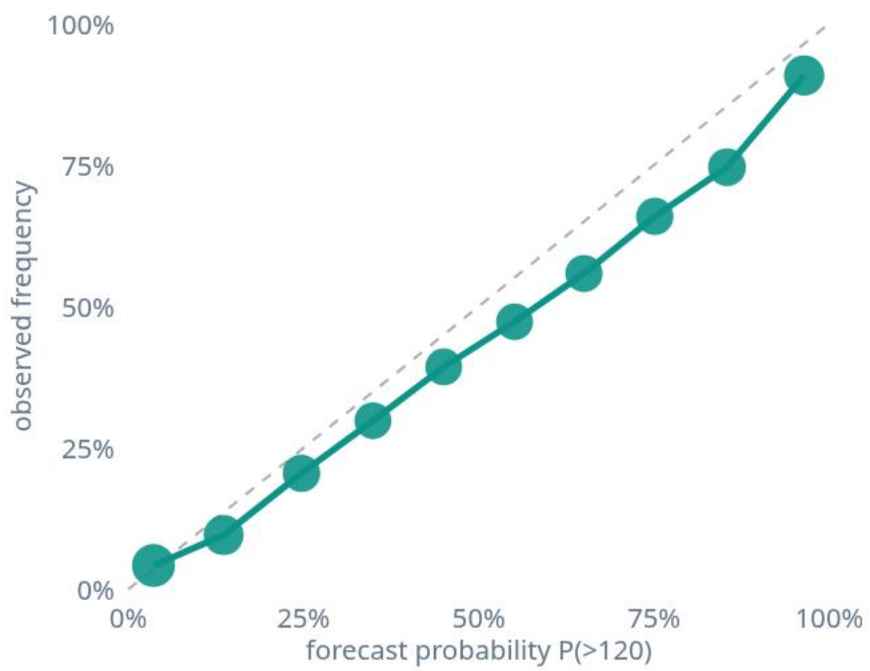
We do not claim to out-model WRF-Chem. Sources: CEEW 'How Well can Delhi Predict Air Quality?' (Ignatious & Rafiuddin, 2025); Ghude et al., GMD 17 (2024); docs/POSITIONING.md

Does it warn before the air turns? Measured on 2025-26 Delhi

horizon	onsets (>120)	onset recall — alarm on P ≥ 0.3	onset recall — persistence	alarm F1 (P ≥ 0.3) / persist.	P(>120) Brier skill	80 % PI coverage
+24 h	15,701	**54%**	0%	0.765 / 0.747	+51%	78%
+48 h	18,226	**54%**	0%	0.744 / 0.708	+46%	78%
+72 h	19,117	**51%**	0%	0.725 / 0.696	+38%	77%

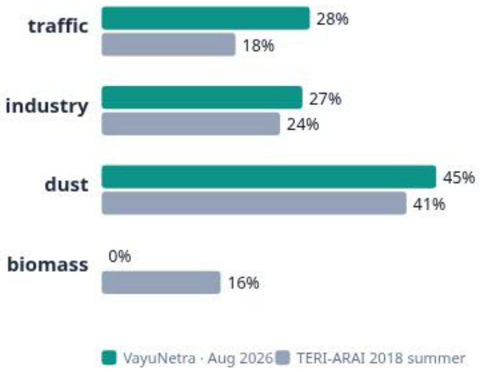
Onset = clean at issue time, above 120 μg/m³ (Very Poor) at t+h. 'Tomorrow = today' has onset recall 0 by construction. **Where we are weak** — the Severe tail (>250): the blended forecast only matches persistence there and the median alarm sees few Severe onsets (as the official WRF-Chem does); the product speaks in calibrated probability: P(>250) Brier skill +31 % at 24 h.

Reliability, P(>120), +24 h

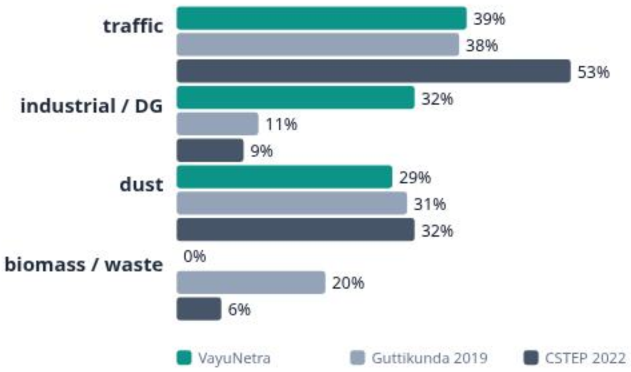


Checked against published apportionment — agreements and disagreements both shown

Delhi · source-only shares



Bengaluru · source-only shares



Who is in the forecast?

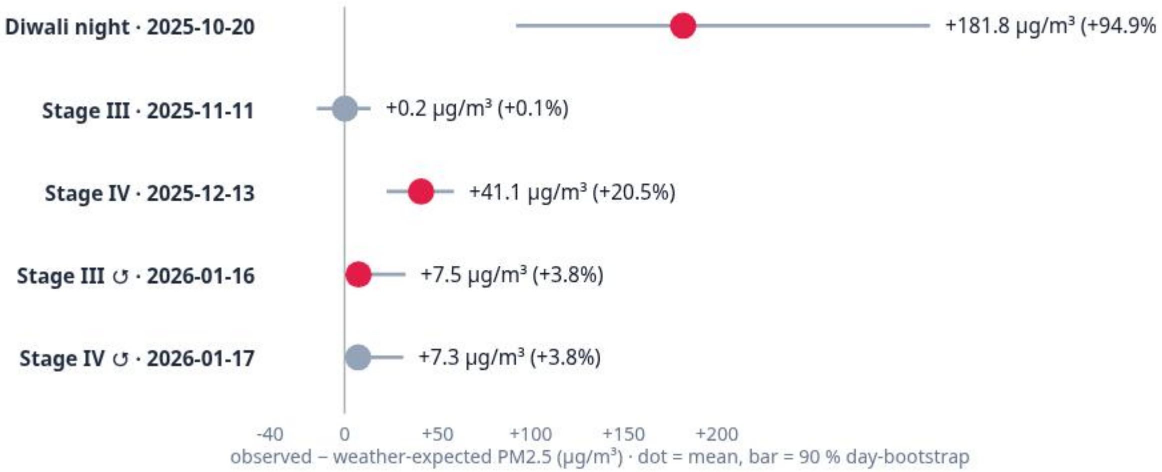
+24 h · Very Poor	**0.1% · ~11,837 people**
+48 h · Very Poor	**0.1% · ~9,980 people**
+72 h · Very Poor	**0.1% · ~19,580 people**

basis: gpw411_cells (1,50,516 people in sampled cells; city-scaled figure assumes the same share city-wide — the card says so)

Σ cell population × calibrated P(> band); GPW v4.11 where sampled, cited city population otherwise. Exposure, not mortality. GET /exposure

Read honestly: rankings agree; we over-read kerbside traffic (arterial-road stations, NO_2/CO signatures) and book less as regional than a dispersion model; our Bengaluru 'industrial' is DG sets, our biomass misses small waste fires. Shares move between daily runs where NO_2 coverage is thin — shown, not smoothed. docs/ATTRIBUTION_VALIDATION.md

Did the air change during the order, weather taken out?



LightGBM on ERA5 meteorology + hour/dow/doy/cell, fitted on 88,270 season hours outside the windows; held-out R^2 0.61, RMSE 65 µg/m³. python -m ml.eval.interventions

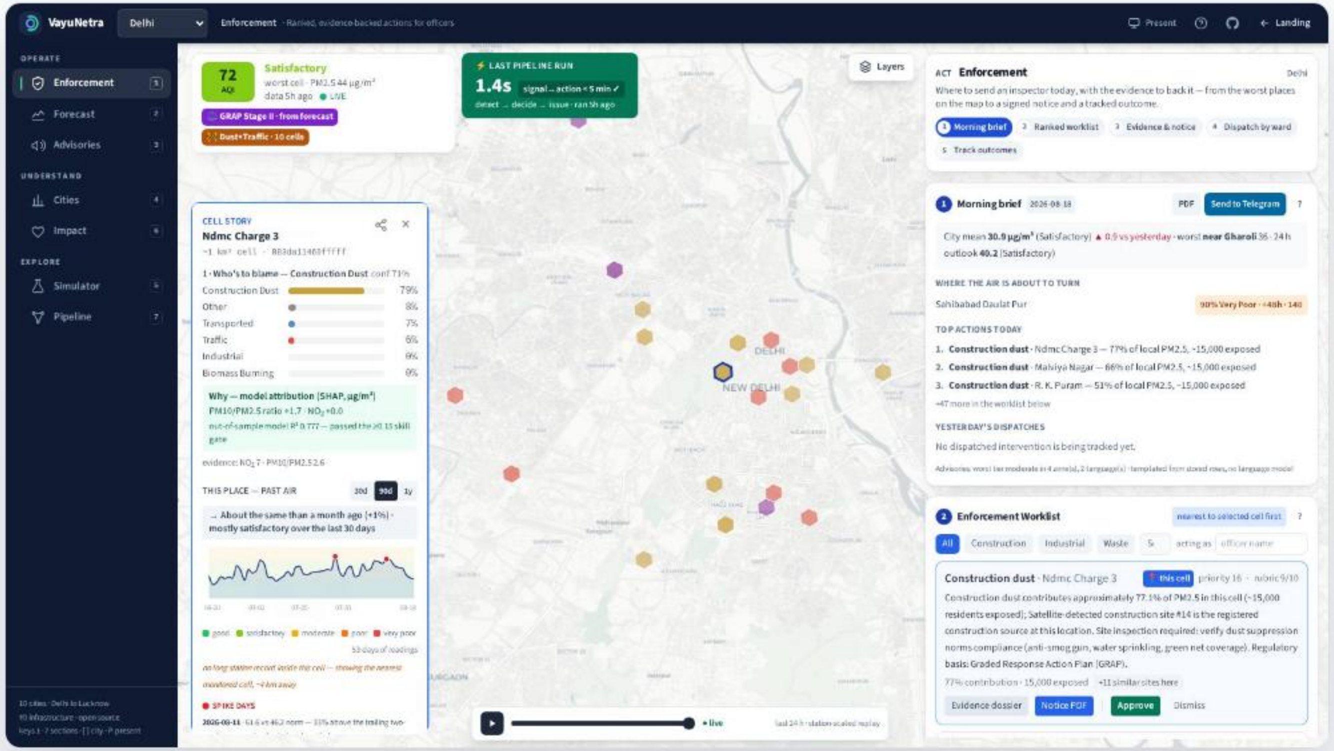
Read honestly

Diwali night shows the method has the power to see a large signal. For the GRAP windows we find **no weather-adjusted reduction this method can detect**; the Stage IV rows sit in the most stagnant weather of the season, where a tree model fitted on calmer hours under-predicts. Association, not causation — coincident factors stay in the number.

This is why VayuNetra measures each dispatched action against **its own cell and the city's drift** — a targeted action has a control; a blanket stage does not.

A14 · THE OFFICER LOOP CLOSES

See → decide → act → prove — with an audit trail



1. **Approve** — moves the item to its ward queue
2. **Dispatch** — freezes the cell's 7-day baseline; effect vs city drift is measured from then on
3. **Close with a finding** — violation found · compliant · inaccessible · n/a, plus a note
4. **History** — immutable log: from → to, who, when, finding; survives the nightly run and even deletion of the record

PRANA-ready CSV — every tracked action with baseline / after / effect / status / closure, mapped to its NCAP head.